# Credit Intelligence Platform: AI-Powered Loan Default Risk Prediction and Decision Support System

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## Problem Statement

- In the banking and financial sector, accurately assessing the credit risk of loan applicants is critical for minimizing financial losses and ensuring responsible lending. Traditional credit evaluation methods are often time-consuming, dependent on manual analysis, and may fail to identify complex risk patterns.
- Financial institutions require a reliable and explainable system that can analyze applicant data, predict loan default probability, and support informed decision-making. An AI-powered credit risk intelligence platform can automate risk assessment, improve prediction accuracy, and provide transparent insights to assist lenders in making faster and more effective credit decisions.

# Objectives

- **Primary Objective:** Develop an AI-powered credit risk intelligence platform to predict loan default risk and support informed lending decisions.
- **Specific Objectives:**
  - Analyze customer financial and credit-related data to identify risk patterns.
  - Predict the probability of loan default using machine learning techniques.
  - Provide explainable insights into model predictions using Explainable AI.
  - Enable users to explore data through a natural language Talk-to-Data interface.
  - Generate business-readable decision rules to support credit policy decisions.
  - Deliver a user-friendly and deployable solution through a web-based interface.

# Proposed System

*"Leveraging AI for Intelligent Credit Risk Assessment"*

- Utilizes machine learning to analyze applicant financial, demographic, and credit-related information.
- Predicts the probability of loan default and categorizes applicants into Low, Medium, or High risk groups.
- Incorporates Explainable AI techniques to provide transparent and interpretable credit decisions.
- Enables natural language data exploration through a Talk-to-Data interface for business users.
- Supports faster, data-driven, and reliable lending decisions while reducing manual assessment efforts.

## Advantages of Proposed System

- **Accurate Risk Assessment:**
  - Predicts loan default probability using machine learning, enabling better credit decisions.
- **Explainable Decision Making:**
  - Provides transparent insights into risk predictions through Explainable AI techniques.
- **Faster Credit Evaluation:**
  - Automates risk analysis, reducing manual effort and decision-making time.
- **Business-Friendly Analytics:**
  - Enables users to explore data through natural language queries and readable business rules.
- **Scalable Deployment:**
  - Dockerized architecture ensures easy deployment and maintenance across environments.

# Software Requirements

- **Programming Language:** Python
- **Machine Learning:** XGBoost, Scikit-Learn
- **Data Processing:** Pandas, NumPy
- **Explainable AI:** SHAP
- **User Interface:** Streamlit
- **Deployment:** Docker, Docker Compose
- **Dataset:** Home Credit Default Risk Dataset (Kaggle)
- **Development Tools:** Visual Studio Code (VS Code)

# Module: Credit Risk Prediction

**Detailed Function:**

- Serves as the core intelligence engine of the platform for assessing applicant creditworthiness.
- Analyzes applicant information such as age, work experience, annual income, loan amount, credit history score, financial stability score, and external credit rating.
- Utilizes the XGBoost Classifier to predict the probability of loan default.
- Learns complex relationships between financial and behavioral attributes from historical credit data.
- Generates a risk score that quantifies the likelihood of repayment failure.
- Supports faster and more consistent lending decisions through automated risk assessment.

**Input:** Applicant financial and credit information.
**Output:** Default probability and risk score.

# Module: Decision Intelligence

**Detailed Function:**

- Transforms model predictions into actionable lending recommendations.
- Evaluates default probability against predefined business thresholds.
- Categorizes applicants into Low Risk, Medium Risk, or High Risk groups.
- Generates approval recommendations such as Approved, Review Required, or Rejected.
- Produces decision insights to assist loan officers in understanding the risk profile.
- Ensures consistency and transparency in credit approval workflows.

**Input:** Default probability generated by the prediction model.
**Output:** Risk category and lending recommendation.

# Module: Explainable AI (SHAP)

**Detailed Function:**

- Provides transparency into the machine learning prediction process.
- Identifies the most influential factors affecting the applicant's risk score.
- Quantifies the positive and negative contribution of each feature.
- Helps financial analysts understand why a particular decision was made.
- Improves regulatory compliance and stakeholder trust through explainable predictions.
- Generates feature importance visualizations and prediction explanations.

**Input:** Applicant features and model prediction results.
**Output:** Feature contribution analysis and explanation report.

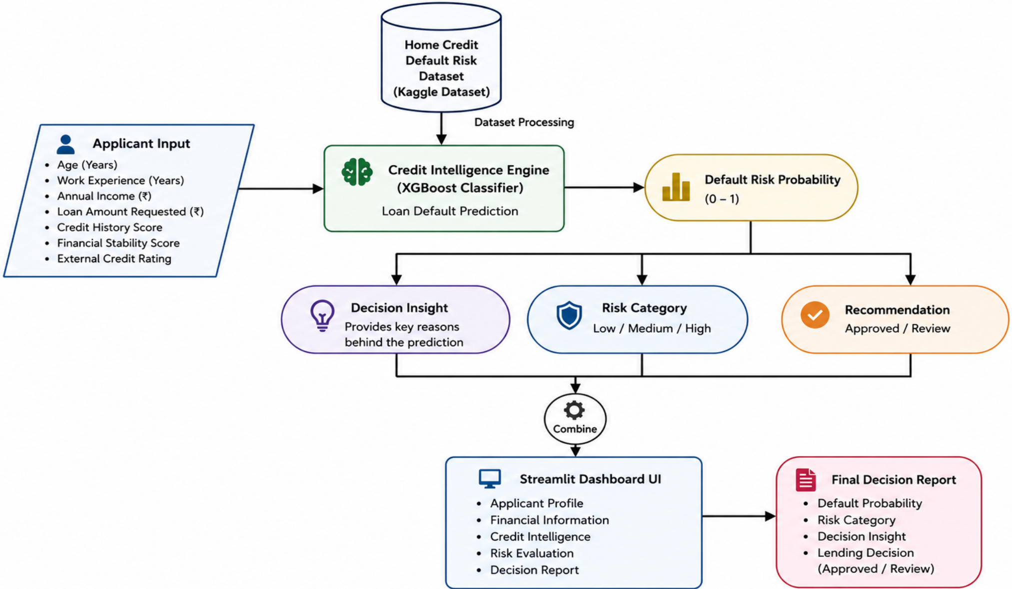
# Module: Dashboard Interface

**Detailed Function:**

- Provides an interactive web-based interface for credit risk evaluation.
- Allows users to enter applicant and financial information through structured forms.
- Displays default probability, risk category, and decision recommendations.
- Presents model outputs in an intuitive and user-friendly manner.
- Supports real-time evaluation and visualization of credit risk.
- Acts as the central platform connecting all system modules.

**Input:** User-entered applicant information.
**Output:** Interactive decision report and risk assessment dashboard.

# Architecture diagram

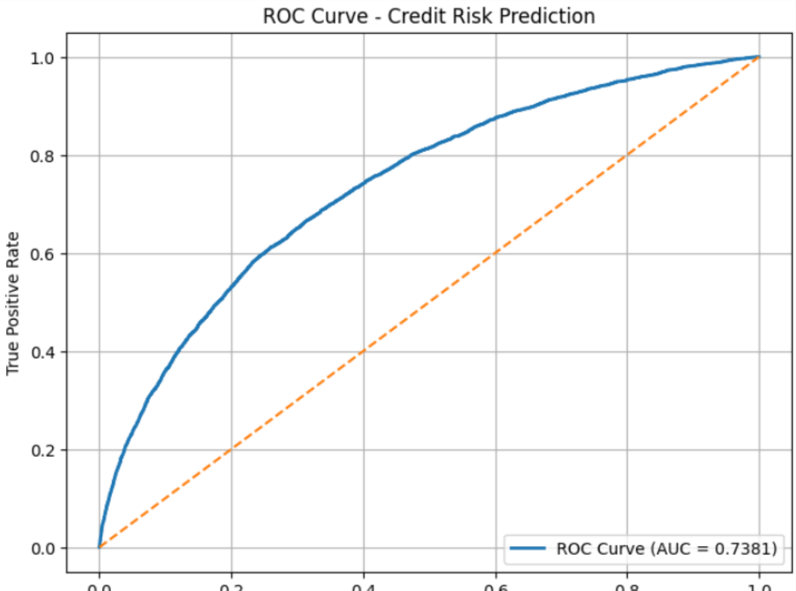


# Results and Discussion

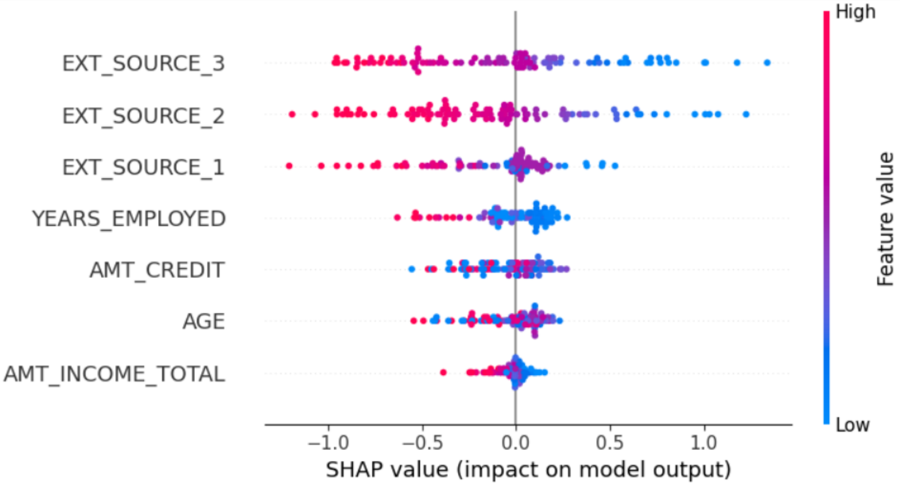
```
===== MODEL PERFORMANCE =====
Accuracy : 0.6925678422190787
Precision: 0.1588785046728972
Recall   : 0.6539778449144008
F1 Score : 0.25564916148334776
ROC AUC  : 0.7380577089255124

===== CLASSIFICATION REPORT =====
              precision    recall  f1-score   support

           0       0.96      0.70      0.81     56538
           1       0.16      0.65      0.26      4965

    accuracy                           0.69     61503
   macro avg       0.56      0.67      0.53     61503
weighted avg       0.89      0.69      0.76     61503


===== CONFUSION MATRIX =====
[[39348 17190]
 [ 1718  3247]]
```

- The XGBoost model achieved strong performance in predicting loan default risk.

# ROC Curve Analysis

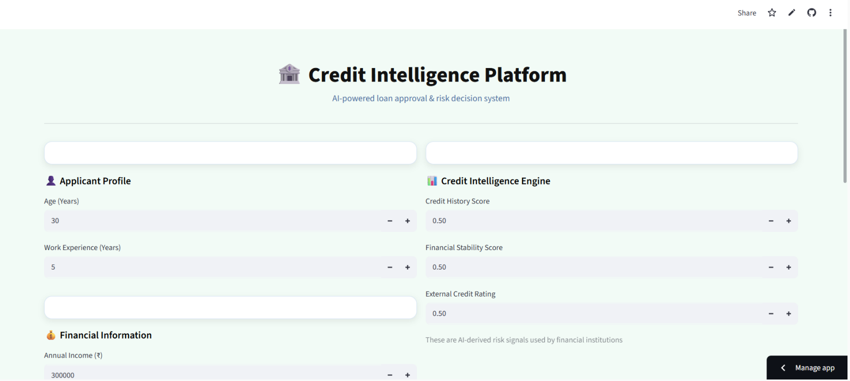


# Explainable AI Results




- SHAP values highlight the most influential factors affecting credit risk predictions.

# Simulation




Click Here to Watch Project Demo

# Conclusion

- **Work Completed So Far:**
  - Analyzed the problem of credit risk assessment and loan default prediction in financial institutions.
  - Performed exploratory data analysis on the Home Credit Default Risk dataset and derived key business insights.
  - Developed a machine learning pipeline using XGBoost for predicting loan default probability.
  - Implemented risk categorization into Low, Medium, and High risk groups.
  - Designed an interactive Streamlit dashboard for real-time credit risk evaluation.
  - Integrated decision insights and recommendation mechanisms to support lending decisions.
  - Containerized the application using Docker for easy deployment and scalability.

# Project Resources

**GitHub Repository:**

https://github.com/Nandana-R-2004/credit-intelligence-platform.git

**Live Application (Streamlit):**

https://credit-intelligence-platform-janfzrqog9appfdu2w4d9xz.streamlit.app/

**Demo Video:**

https://drive.google.com/file/d/1ar2_FZyCNEdtb6Th8N4VROg1H_kHVKL3/view?usp=sharing

**Thank You!**

**Questions?**